

DAY 20: STATISTICAL INFERENCE & SAMPLING

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60-Day Data Science Learning Journey

Goal: Learn how samples are used to understand populations and quantify uncertainty with sampling distributions, standard error, confidence intervals and hypothesis testing.

Prerequisite: Day 19 - Probability Distributions

1. Population vs Sample

Population: the complete group we want to understand.
Sample: a subset selected from that population.
Example: A university has 10,000 students. All 10,000 are the population. A selected group of 200 students is the sample.

2. Why Use Samples?

Studying an entire population can be expensive, slow or impractical. A carefully designed sample can provide useful information about the larger population.

3. Sampling Methods

Simple random sampling: every member has a chance of selection.

```
import numpy as np

students = np.arange(1, 101)

sample = np.random.choice(
    students,
    size=10,
    replace=False
)

print(sample)
```

With **replace=True**, an item can be selected again. With **replace=False**, selected items cannot repeat.

4. Sampling Bias

Sampling bias occurs when the sampling process systematically produces a sample that does not represent the target population well. Examples include convenience sampling, undercoverage, non-response and voluntary-response bias.

5. Sample Mean

```
import numpy as np

marks = [70, 75, 80, 85, 90]

sample_mean = np.mean(marks)

print(sample_mean)
```

Output: **80.0**. The sample mean is a statistic and can be used as an estimate of a population mean.

6. Sampling Distribution

If we repeatedly take samples and calculate a statistic such as the mean each time, the distribution of those statistics is called a sampling distribution.

```
sample_means = []

for i in range(1000):
    sample = np.random.choice(
        marks,
        size=3,
        replace=True
    )
    sample_means.append(np.mean(sample))
```

```
print(sample_means[:10])
```

The exact values vary because the sampling is random.

7. Standard Error

Standard deviation describes variation among observations. Standard error describes sampling variation of a statistic such as a sample mean. A common estimate is $SE = s / \sqrt{n}$.

```
import numpy as np

sample_std = 10
sample_size = 100

standard_error = (
    sample_std / np.sqrt(sample_size)
)

print(standard_error)
```

Output: 1.0 . Larger samples generally produce smaller standard errors.

8. Central Limit Theorem

Under appropriate conditions, the distribution of sample means tends toward an approximately Normal distribution as sample size becomes sufficiently large, even when the underlying population is not Normal.

9. Point Estimation

A point estimate is a single value used to estimate a population parameter. The sample mean is a common point estimate for the population mean.

```
sample = [72, 75, 80, 83, 90]

estimate = np.mean(sample)

print(estimate)
```

Output: 80.0 .

10. Confidence Intervals

A confidence interval is an interval produced by a statistical procedure to express uncertainty around an estimate. For a mean with known population standard deviation, a common 95% Normal-based interval is $\bar{x} \pm 1.96(\sigma/\sqrt{n})$.

```
import numpy as np

mean = 80
std = 10
n = 100

se = std / np.sqrt(n)
margin = 1.96 * se

lower = mean - margin
upper = mean + margin

print("Lower:", lower)
print("Upper:", upper)
```

Output: Lower = 78.04 , Upper = 81.96 .

Important: A 95% confidence level refers to the long-run coverage of the procedure under its assumptions. It is not correct to interpret the completed interval as having a 95% probability of containing a fixed parameter.

11. Confidence Level and Margin of Error

Common confidence levels are 90%, 95% and 99%. Higher confidence generally gives a wider interval, all else equal. The margin of error is the distance from the estimate to an interval endpoint.

Confidence	Common z critical
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90%	1.645
95%	1.96
99%	2.576

12. Hypothesis Testing

Hypothesis testing provides a formal procedure for evaluating a claim about a population using sample data.

Null hypothesis (H_0): the baseline claim tested by the procedure.
Alternative hypothesis (H_a): the competing claim.

Example: $H_0: \mu = 70$ and $H_a: \mu \neq 70$ for a two-sided test.

13. P-Value

The p-value is the probability, assuming the null hypothesis is true, of obtaining a result at least as extreme as the observed result, according to the specified test.

With $\alpha = 0.05$, $p < 0.05$ provides evidence against H_0 under the test. If $p \geq 0.05$, there is insufficient evidence to reject H_0 . This does not prove H_0 is true.

14. Type I and Type II Errors

Type I: rejecting a true null hypothesis, often called a false positive.
Type II: failing to reject a false null hypothesis, often called a false negative.

15. One-Sample t-Test in Python

```
from scipy import stats

marks = [72, 75, 78, 80, 82, 85, 88, 90]

result = stats.ttest_1samp(
    marks,
    popmean=75
)

print("T-statistic:", result.statistic)
print("P-value:", result.pvalue)
```

This performs a two-sided one-sample t-test by default. H_0 states that the population mean equals 75.

16. Day 20 Practice Tasks

1. Create a NumPy population containing 1 to 100 and select a random sample of 10.
2. Calculate the sample mean.
3. Generate 1,000 sample means using samples of size 10 and calculate their mean.
4. Calculate standard error when sample standard deviation = 12 and sample size = 144.
5. Calculate a 95% Normal-based confidence interval when mean = 75, population standard deviation = 10 and n = 100.
6. Explain Population, Sample, Parameter and Statistic.
7. Explain Type I and Type II errors.
8. Perform a one-sample t-test on [72,75,78,80,82,85,88,90] against population mean 75.
9. Simulate 1,000 sample means from student marks.
10. Build a complete Student Statistical Inference Analyzer.

17. Complete Practice Solutions

```
import numpy as np

population = np.arange(1, 101)

sample = np.random.choice(
    population,
    size=10,
    replace=False
)

print(sample)
```

Explanation: The sample contains 10 unique values because replace=False.

```
sample_mean = np.mean(sample)
print(sample_mean)
```

Explanation: This calculates the arithmetic mean of the selected sample.

```
sample_means = []

for i in range(1000):
    sample = np.random.choice(
        population,
        size=10,
        replace=False
    )
    sample_means.append(np.mean(sample))

print(np.mean(sample_means))
```

Explanation: The average of many sample means should be close to the population mean, with random variation.

```
import numpy as np
se = 12 / np.sqrt(144)
print(se)
```

Explanation: Output: 1.0.

```
mean = 75
std = 10
n = 100

se = std / np.sqrt(n)
margin = 1.96 * se

print(mean - margin)
print(mean + margin)
```

Explanation: Output: 73.04 and 76.96.

```
from scipy import stats

marks = [72, 75, 78, 80, 82, 85, 88, 90]

result = stats.ttest_1samp(
```

```

marks,
popmean=75
)

print(result.statistic)
print(result.pvalue)

```

Explanation: SciPy returns the t-statistic and p-value for the specified two-sided one-sample t-test.

18. Full Student Statistical Inference Analyzer

```

import pandas as pd
import numpy as np
from scipy import stats

data = {
    "Student": [
        "Ali", "Usman", "Hamza", "Ahmad",
        "Bilal", "Hassan", "Zain", "Omar",
        "Ayan", "Saad"
    ],
    "Marks": [
        72, 75, 78, 80, 82,
        85, 88, 90, 76, 84
    ]
}

df = pd.DataFrame(data)

mean_marks = df["Marks"].mean()
std_marks = df["Marks"].std()
n = len(df)

standard_error = std_marks / np.sqrt(n)

margin_of_error = 1.96 * standard_error

lower = mean_marks - margin_of_error
upper = mean_marks + margin_of_error

test = stats.ttest_1samp(
    df["Marks"],
    popmean=75
)

print("Average Marks:", mean_marks)
print("Standard Deviation:", std_marks)
print("Sample Size:", n)
print("Standard Error:", standard_error)

print("\n95% Confidence Interval:")
print("Lower:", lower)
print("Upper:", upper)

print("\nT-statistic:", test.statistic)
print("P-value:", test.pvalue)

print("\nAbove Average Students:")
print(df[df["Marks"] > mean_marks])

```

Line-by-line explanation

1. Import Pandas for tabular data.
2. Import NumPy for numerical calculations.
3. Import SciPy statistics tools.
4. Create the student dataset.
5. Build a DataFrame.
6. Calculate the sample mean.
7. Calculate sample standard deviation. Pandas std() uses the sample convention by default.
8. Store the sample size.
9. Calculate standard error.
10. Calculate the 95% margin of error.
11. Build the lower and upper interval limits.
12. Run a one-sample t-test against 75.
13. Print the inference results.
14. Filter students above the sample mean.

19. Important Distinctions

Term	Meaning	Example
Population	Entire target group	All university students

Sample	Subset of population	200 selected students
Parameter	Population characteristic	Population mean μ
Statistic	Sample characteristic	Sample mean $\bar{x}$

20. Common Mistakes

- Treating a biased sample as representative.
- Confusing a statistic with a parameter.
- Thinking a larger sample automatically removes all bias.
- Misinterpreting a confidence interval as a probability statement about a fixed parameter.
- Saying $p \geq 0.05$ proves H_0 .
- Forgetting that statistical significance does not automatically mean practical importance.
- Ignoring assumptions behind a statistical test.

21. Quick Revision

Population: complete target group.
Sample: selected subset.
Statistic: number calculated from a sample.
Parameter: number describing a population.
Sampling distribution: distribution of a statistic over repeated samples.
Standard error: sampling variability of a statistic.
Confidence interval: interval produced by an inference procedure.
 H_0 : null hypothesis.
 H_A : alternative hypothesis.
p-value: evidence measure under H_0 for the specified test.

22. Final Checklist

■ Population and sample
 ■ Random sampling
 ■ Sampling bias
 ■ Sampling distributions
 ■ Standard error
 ■ Central Limit Theorem
 ■ Point estimation
 ■ Confidence intervals
 ■ Null and alternative hypotheses
 ■ P-values and significance levels
 ■ Type I and Type II errors
 ■ One-sample t-test

23. Day 20 Takeaway

Today you learned how Data Scientists move from sample data toward population-level inference. Sampling, standard error, confidence intervals and hypothesis testing form an important bridge between descriptive statistics and advanced statistical learning.